Data Governance, Compliance and Ethics

ETHICAL ISSUES PERTAINING TO DATA (PART 2)



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Overview of course content

1. Data Governance and Management (Weeks 1-3)
2. Privacy and Data Protection (Weeks 4-6)
3. Ethical Issues Pertaining to Data (Weeks 7-9)
4. Fairness, Accountability, and Transparency of Algorithmic Systems (Weeks 10,11)

Ethical Issues Pertaining to Data (Part 2)

Ethical Issues Pertaining to Data

Ethics and Computing – examining moral problems when using the Internet - spam, censorship and free speech, anonymity offered by the Internet.

Ethical issues arising from the increasing use and pervasiveness of Information Technology and socio-technical systems.

Health technology.. Pervasive monitoring and tracking.. Image, video and sound capture.. Identity.. Perpetuity of data storage.. Transnationality..

Recap

In Week 6, we looked at

- Nature and source of ethics
  - Morality, Ethics and Law
  - Fundamental ethical values
  - Context for Business and IT ethics
- Focussed on a number of issues relating from the use of the internet and ethics
  - Spam, Censorship and free speech, Anonymity, Cyberbullying, Online reputation management and Copyright
- Ethical rules for professional computer users (ACM)

Principle based ethics

This week, we build on this to consider ethics for data systems based on first principles.

We know that it is unlikely that any set of rules can cover every eventuality especially given the pace at which technology evolves. To do this we consider

- The application of Ethics as part of a broad perspective of IT as being part of a broader "Socio-technical" system particularly in related to how we might approach an **ethics impact assessment**
- The application of ethics as part of a **broader research approach**. (This will be particularly relevant to the course project that asks students to review these issues in relation to their own work).

Recall: "the context for IT ethics"

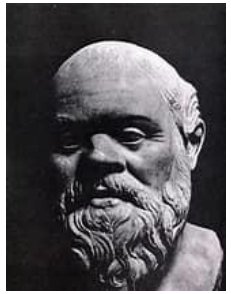
"…the use of information technology in society is creating a rather unique set of ethical issues that requires the making of new moral choices on the part of society and has spawned special implications for its members. Technology itself is not the only, nor necessarily the most responsible, cause of these issues. All ethical questions arise initially out of human agency. Technology, due to its capability to augment mental and physical powers of human beings, does stand in the role of co-conspirator. The lure of power-enhancing capabilities makes technology an inducer of sorts, a necessary but not sufficient underpinning to many of the ethical issues we face today."

Mason "Applying Ethics to Information Technology Issues" (1995) 38 Communications of the ACM 55

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"This discovery of yours will create forgetfulness in the learners' souls, because they will not use their memories; They will trust the external written characters and not remember of themselves."



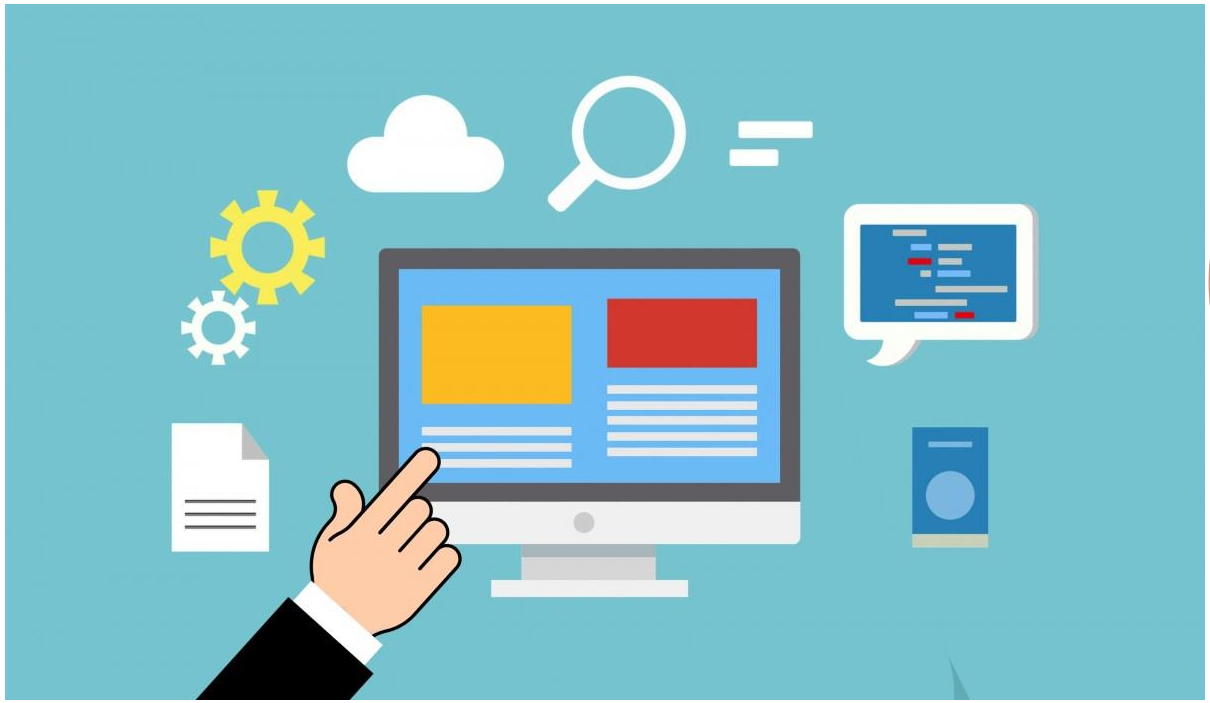
Socrates (469-399 B.C.) on the latest Greek technology – Writing!

Mason "Applying Ethics to Information Technology Issues" (1995) 38 Communications of the ACM 55

Ethics and Computing

- **Computer ethics** focuses on human activities and endeavors influenced or directed by use of computer technologies
- Computers have an increasingly commonplace influence on decision-making processes assumed by most individuals.
- Computer technology is a rapidly developing field which makes it difficult to predict what is coming next so it is essential to have a first principles based approach to computer ethics
- *"Sending an e-mail, writing a program script, creating a database, generating graphics, designing software, processing information and buying software are such behaviors carrying ethical problems within their natures."* (Source: Ethical Issues in Computer use: A Study from Islamic Perspective (core.ac.uk))

So, it's "just grabbing the data" (?)



"For one computer scientist mining data for aggregation and data modelling, data was generally thought of as usable without the need for any form of permission: 'the idea of just grabbing the data, maybe six months, maybe potentially years after it was written, I don't think that requires ethical approval or necessarily the consent of the people involved'" (researcher 9)

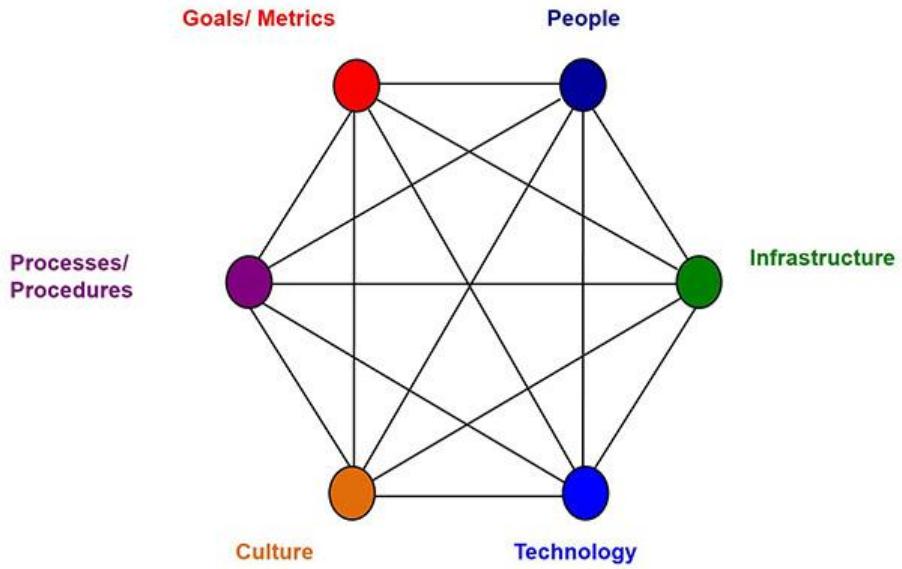
Source: Samuel, G., Derrick, G. E. and van Leeuwen, T. (2019) 'The Ethics Ecosystem: Personal Ethics, Network Governance and Regulating Actors Governing the Use of Social Media Research Data', *Minerva*. Springer Netherlands, 57(3), pp. 317–343. doi: 10.1007/s11024-019-09368-3.

Information Technology and socio-technical systems

An Information technology system requires the use of computers to store, retrieve, transmit, and manipulate data or information.

A **socio-technical system** (STS) is one that considers requirements spanning hardware, software, personal, and community aspects.

It implies an understanding of the **social** structures, roles and rights (from the legal and **social** sciences) to inform the design of **systems** that involve communities of people and **technology**.



Implications of taking a wider STS view

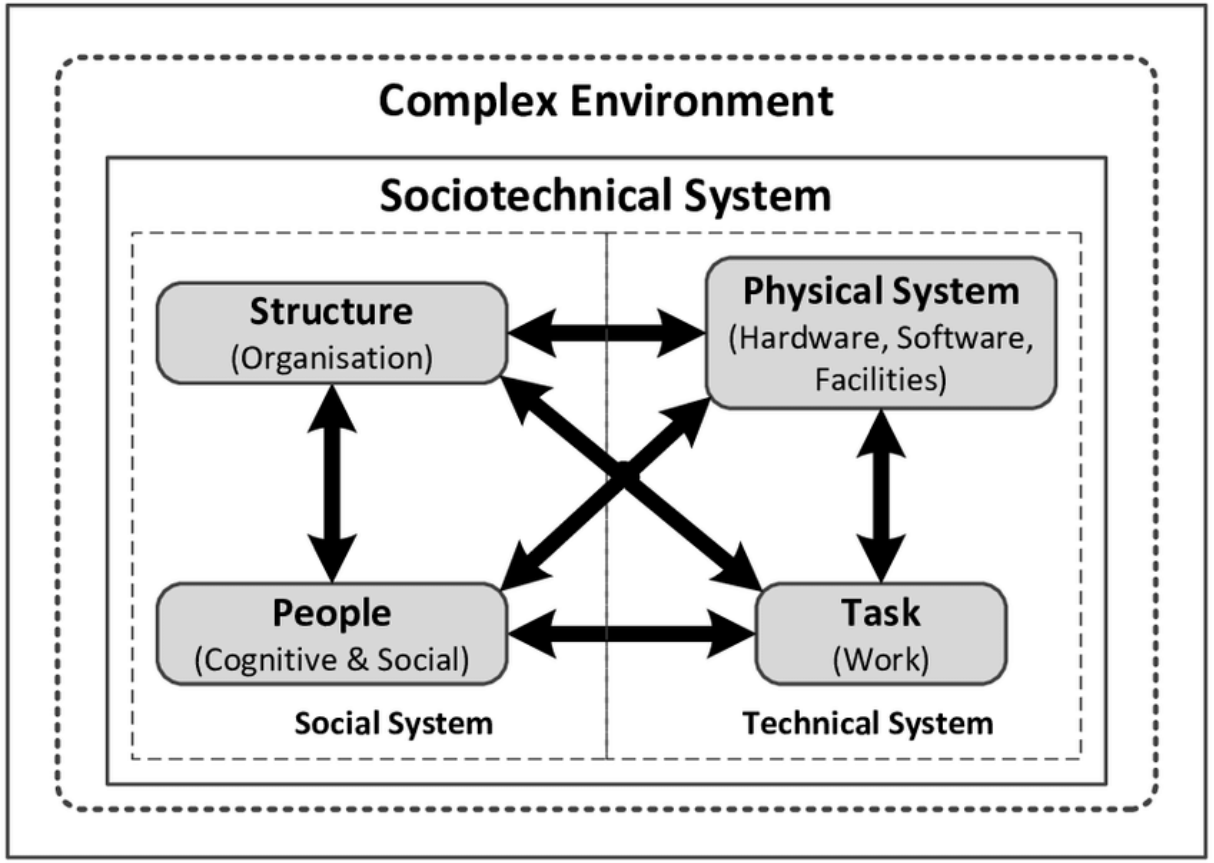
Socio-technical systems: approach in organizational development to examining complex organizational work design that recognizes the interaction between people and technology in workplaces. The term also refers to the broader interaction between society's complex infrastructures and human behaviour.

Socio-technical theory has at its core, the idea that the design and performance of any organisational system can only be understood and improved if both 'social' and 'technical' aspects are brought together and treated as interdependent parts of a complex system.

Components of Socio-technical Systems

Technical system
- Technology
- Business processes and tasks

Social system
- Management practices
- Organisation culture
- People (cognitive and social behaviour)





Socio-Technical Systems Example

STS are systems that include technical systems but also operational processes and people who use and interact with the technical system. Socio-technical systems are governed by organizational policies and rules.

Example

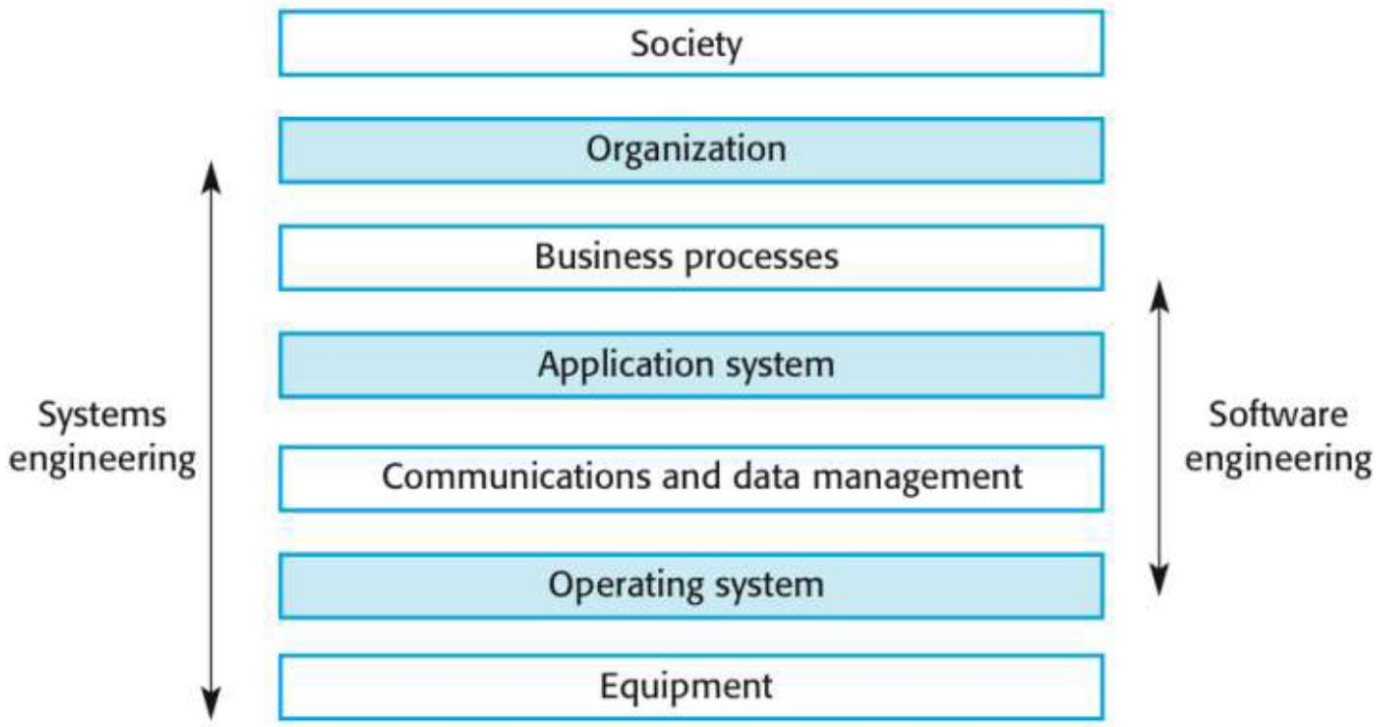
- Software engineers in Silicon Valley may build a software with all sorts of bells and whistles expecting everyone to be tech savy.
- People are in fact elderly and unaccustomed to the interface
- That's the reason we are interested not only in technical dimension but also domain of Socio-technical systems

Socio-technical Systems Stack

Technology stack concept is normally addresses hardware, communications, middleware and software components

We add business/organisational and societal layers both with clear societal links

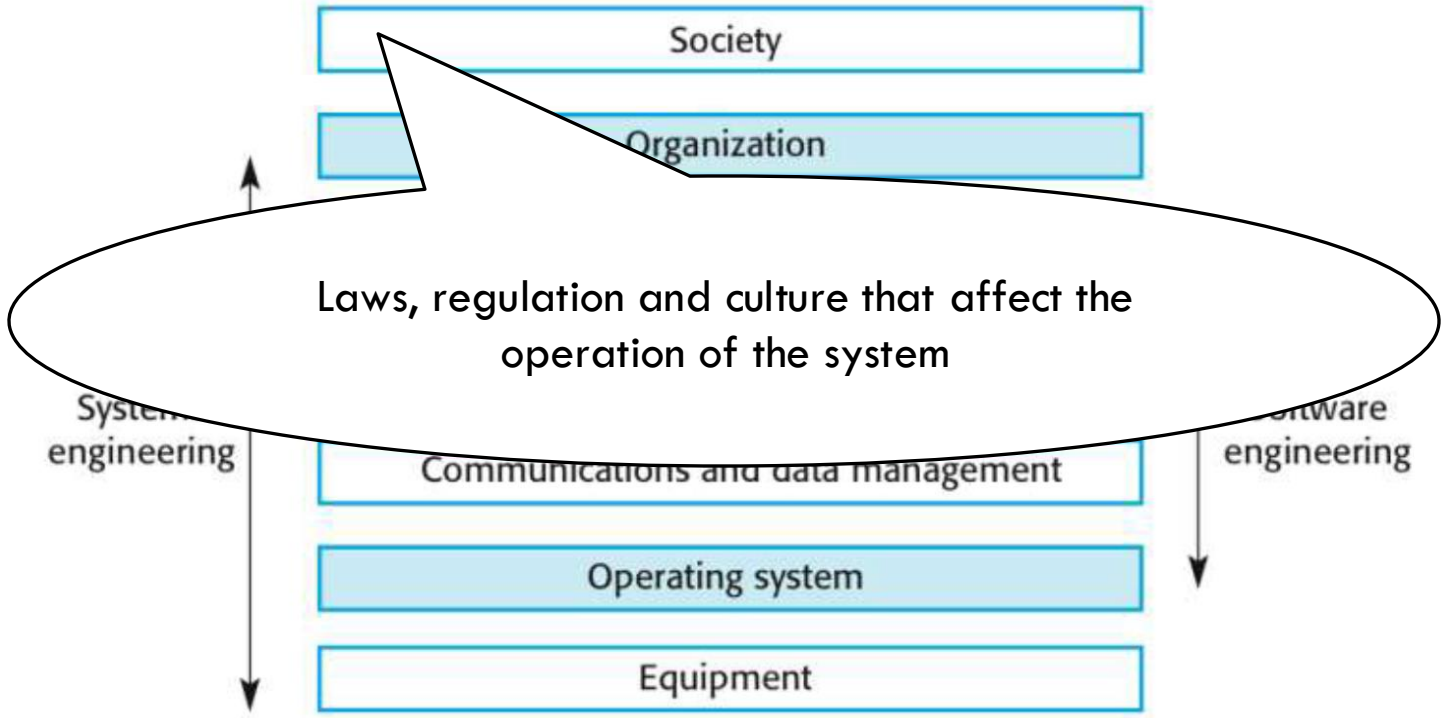
Here we go down the 7 layers looking for potential links between social and technical choices



Socio-technical Systems Stack (1)

The societal layer:
Laws, regulations and culture that govern the operation of the system are defined.

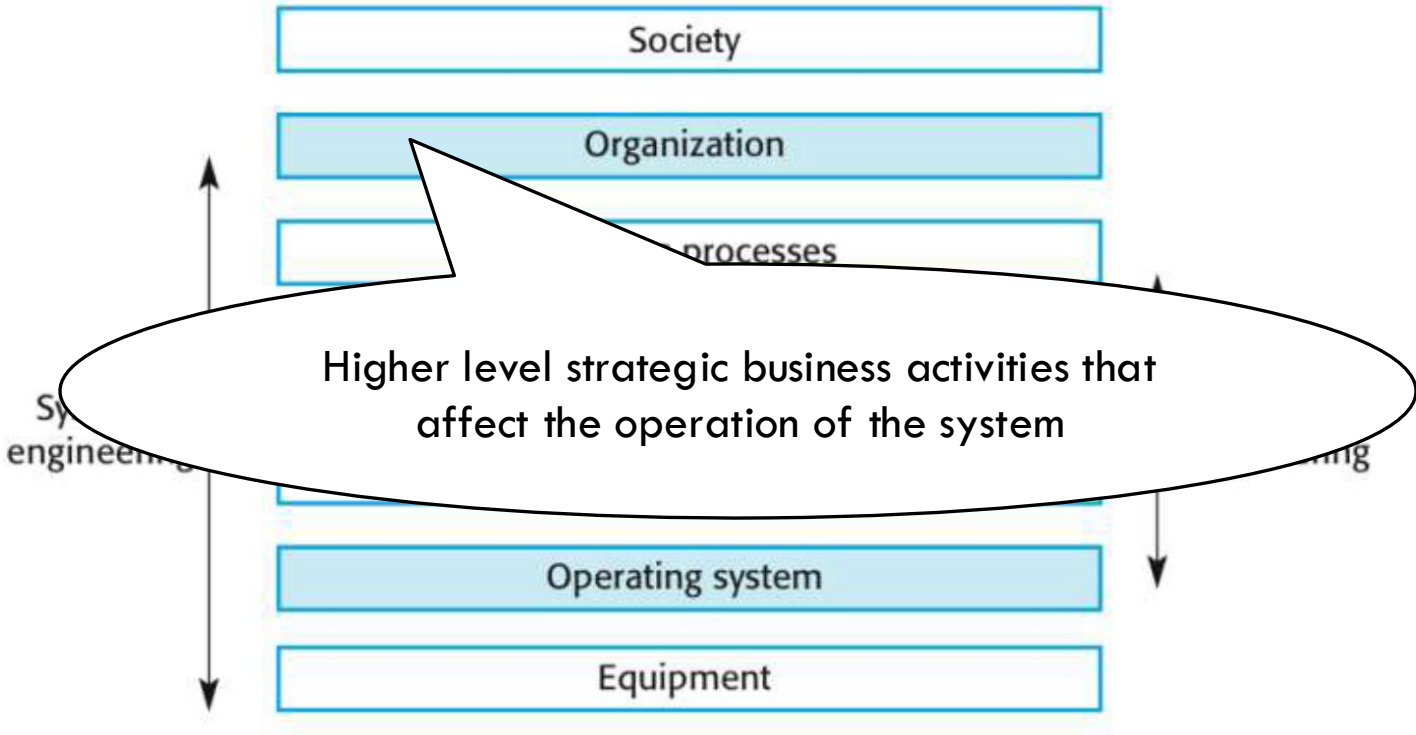
Example: social attitudes to privacy, data protection legislation



Socio-technical Systems Stack (2)

The organizational layer:
At this level, the business rules, regulations, policies along with high-level strategic processes are defined and are to be followed when using the system.

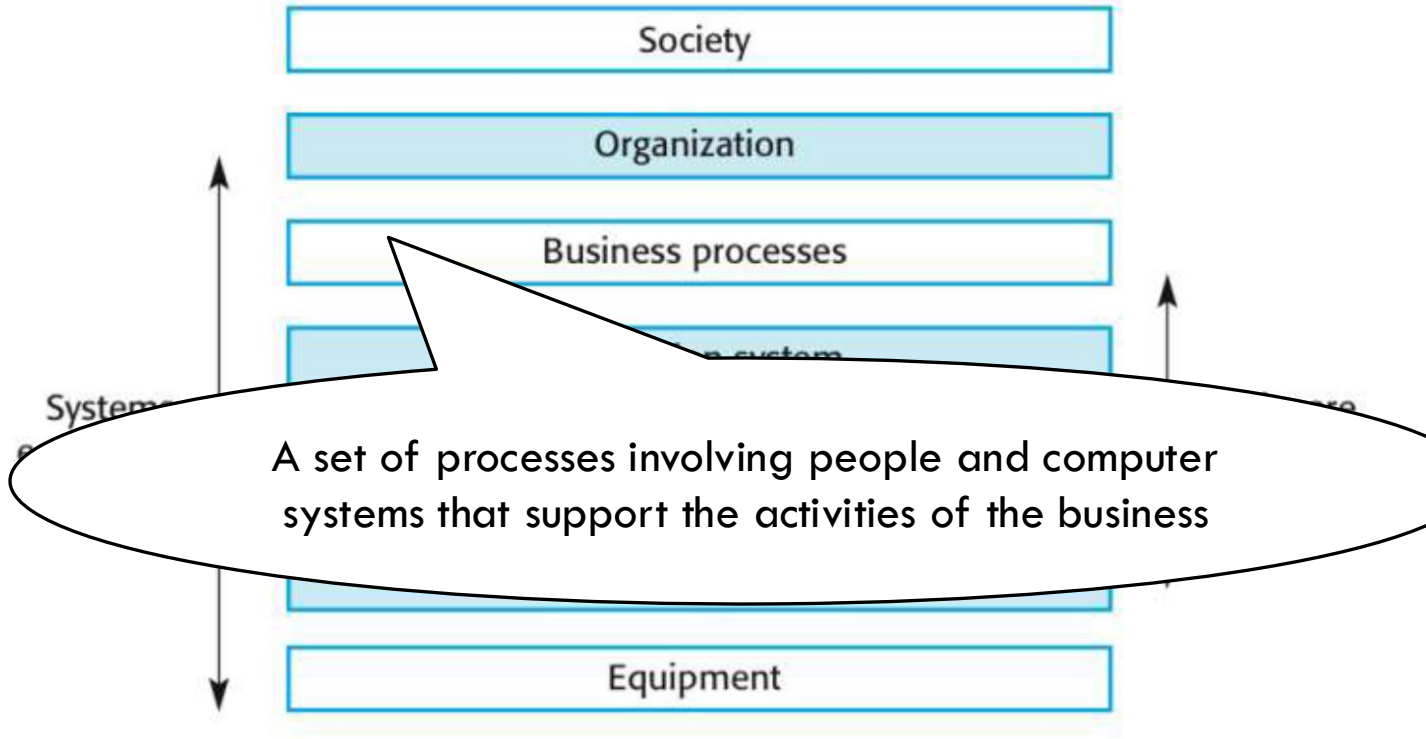
Example: "tone from the top", Data Governance regime



Socio-technical Systems Stack (3)

The business process layer: This layer consists a set of processes involving people and computer systems that support the activities of the business. The use of software system, are defined and enacted.

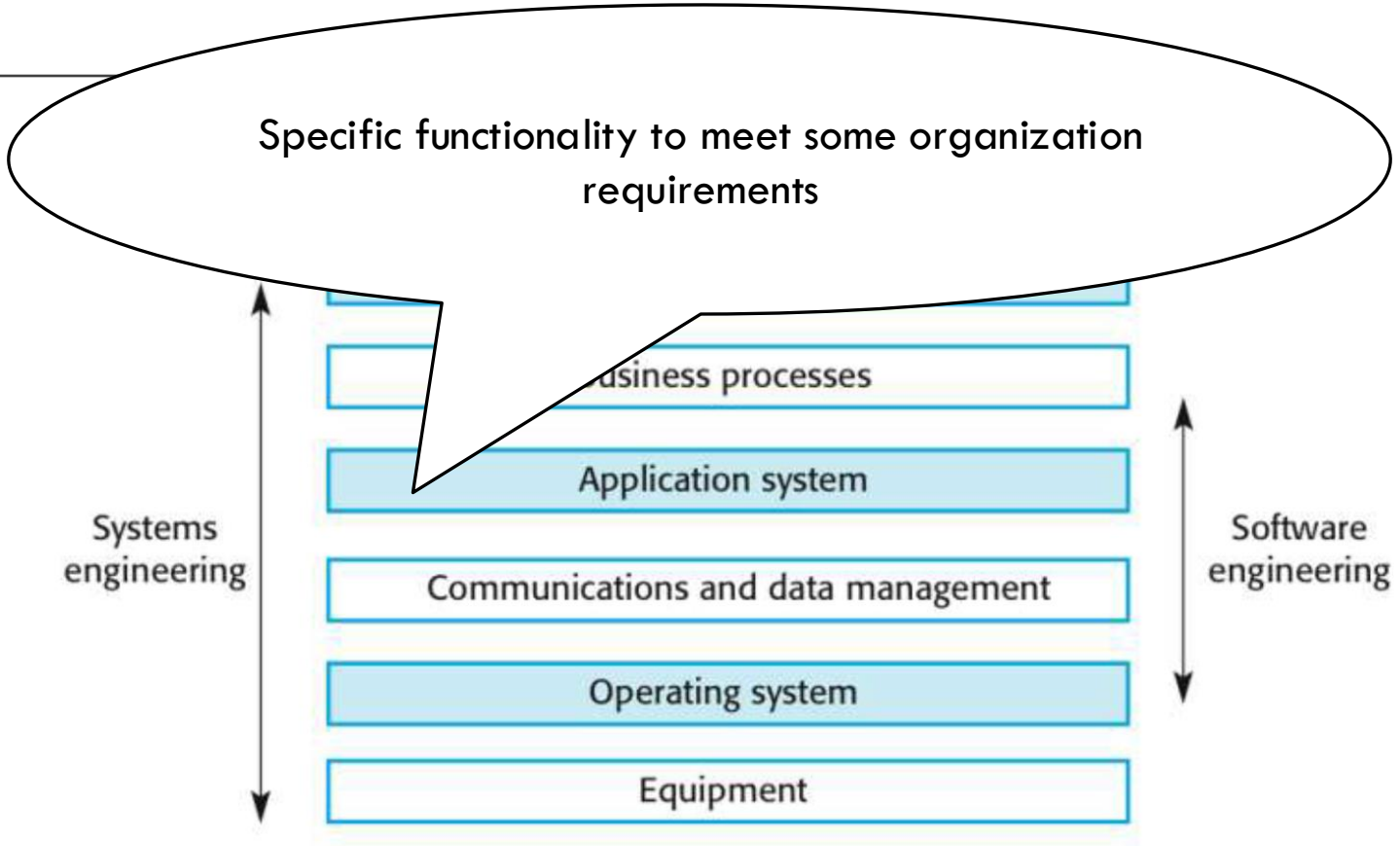
Example: access controls, level of auditing



Socio-technical Systems Stack (4)

The application layer:
This layer provides more specific functionality to meet some organization requirements. There may be many different application programs in this layer.

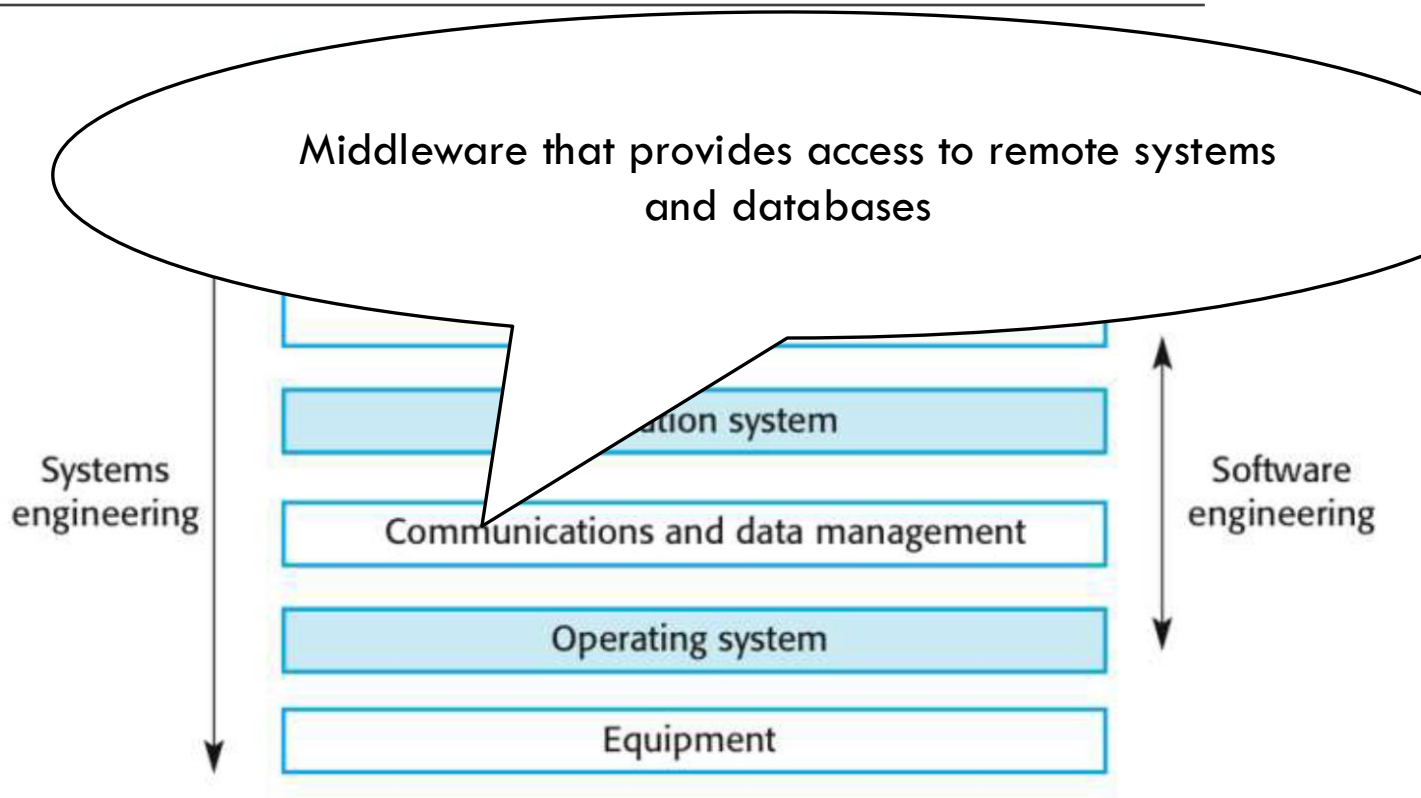
Example: Application-level controls, authentication methods



Socio-technical Systems Stack (5)

The communications and data management layer: extends the operating system facilities and provides an interface that allows interaction with more extensive functionality, such as access to remote systems, access to a system database, etc. This is sometimes called middleware, as it is in between the application and the operating system.

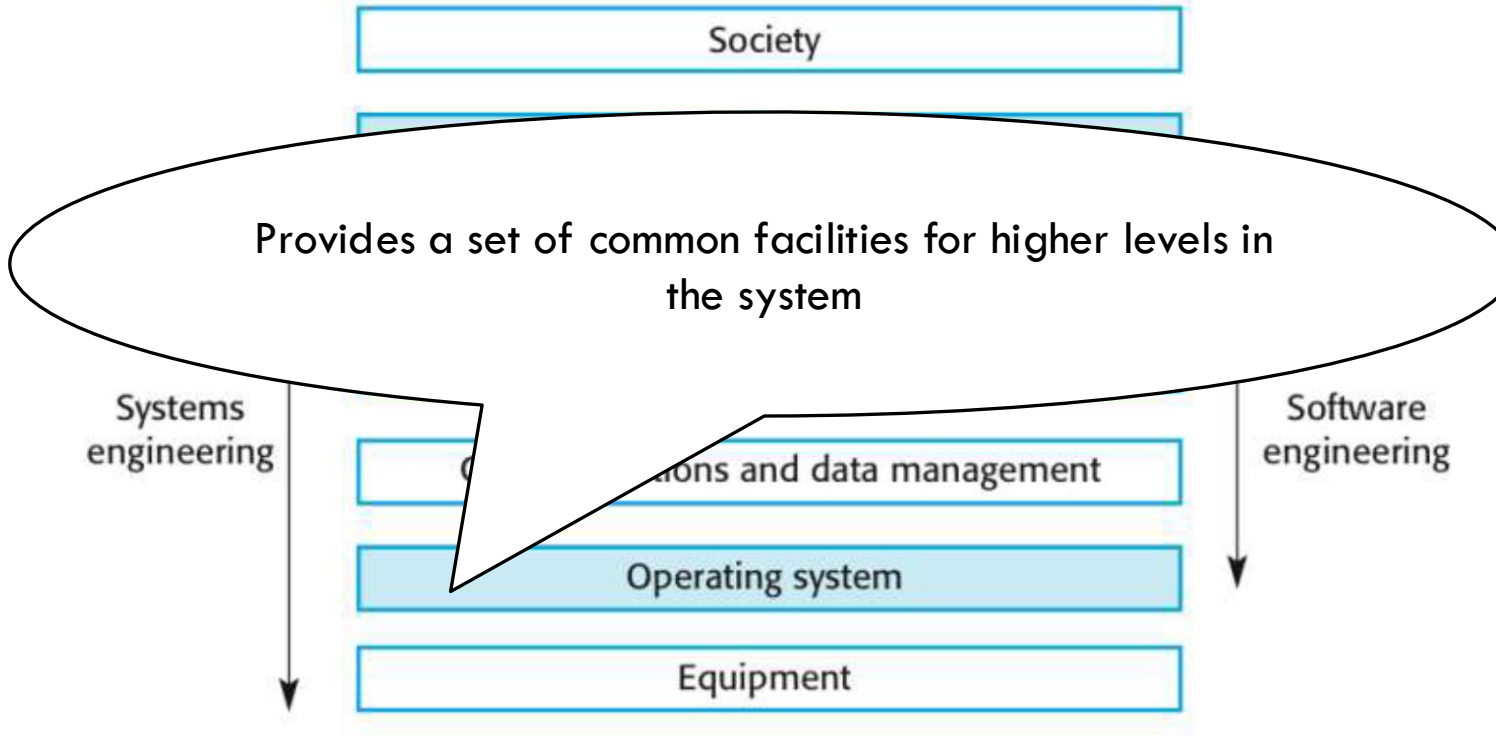
Example: security protocols and degree of distribution



Socio-technical Systems Stack (6)

The operating system layer: This layer provides a set of common facilities for higher software layers in the system. This layer acts as a bridge to the hardware as it allows interaction between software and hardware.

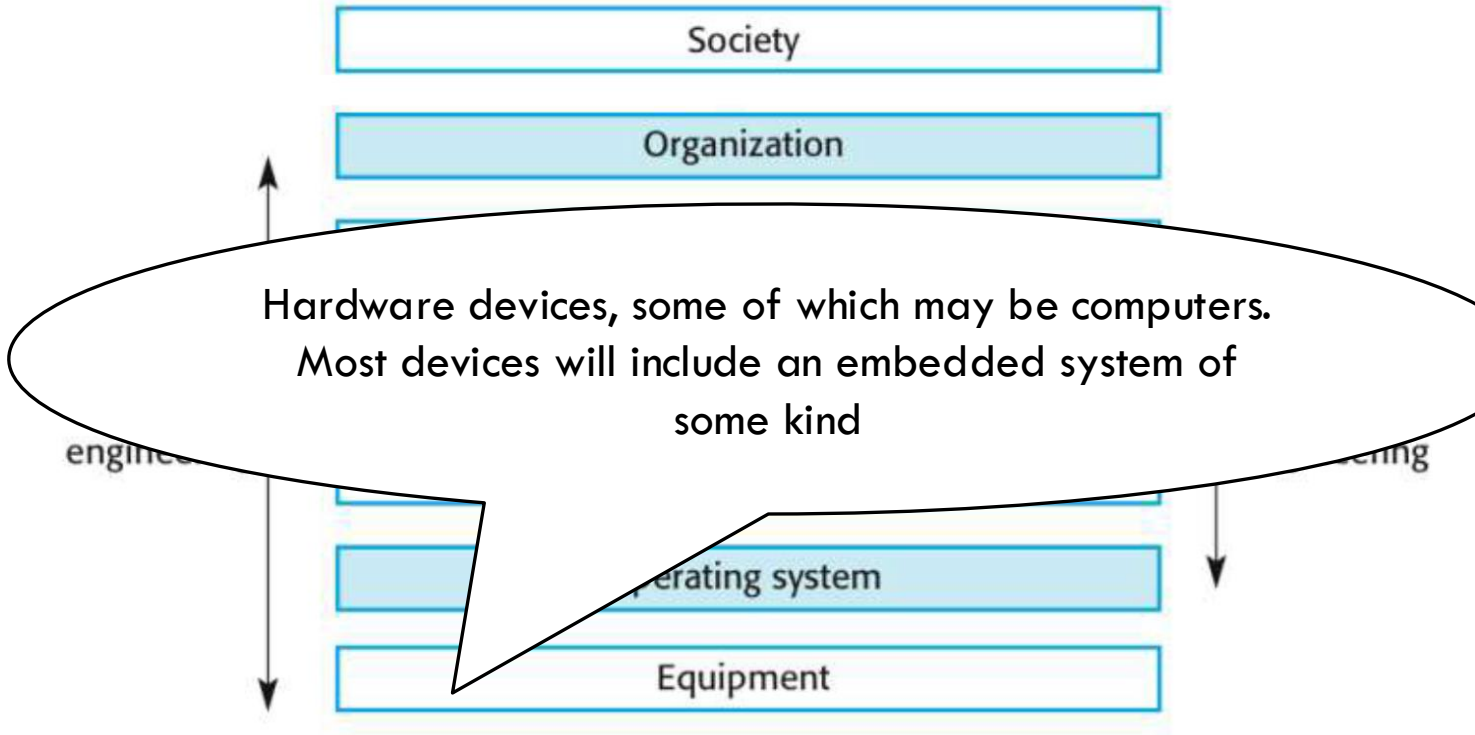
Example: choice of proprietorial or open systems



Socio-technical Systems Stack (7)

The equipment layer:
It contains set of hardware devices some of which may be computer, laptops, phones, etc. Most of the devices include embedded system of some kind.

Example: Hardware configuration, degree of mobile use



Deterministic, non-deterministic and emerging characteristics of STS

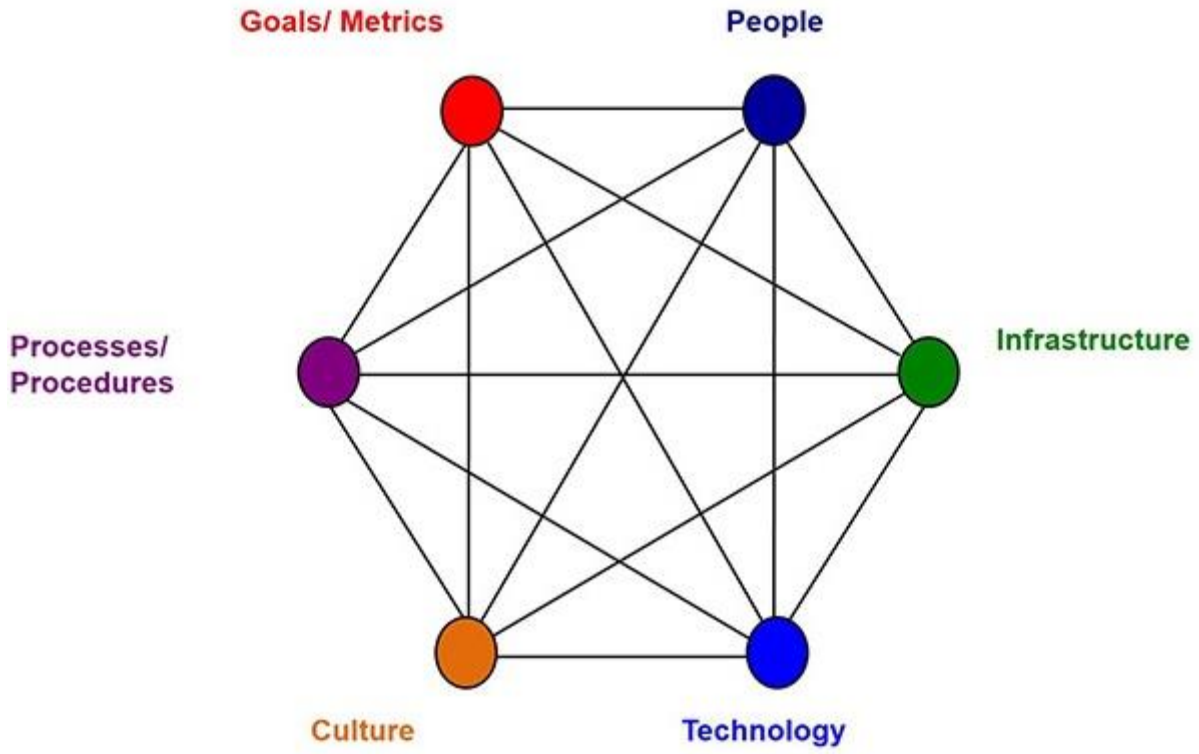
- **Deterministic system**: one where a given sequence of inputs will always produce the same sequence of outputs. Software systems are deterministic.
- **Non-deterministic system:** one which do not always produce the same output when presented with the same input because the system's behaviour is partially dependent on non-deterministic factors such as human operators who do not always behave in the same way for a given set of inputs.
- **Emergent properties:** Properties of the system of a whole that depend on the system components and their relationships. Emergent properties are a consequence of the relationships between system components. They can therefore only be assessed and measured once the components have been integrated into a system.
- **A socio-technical system contains both deterministic and non-deterministic elements.** It will not always produce the same sequence of outputs from the same input sequence. System behaviour is unpredictable because of frequent changes to human and system elements (hardware, software and data).
- **A STS therefore has a complex relationships with organisational objectives.** The extent to which the system supports organizational objectives does not just depend on technical design of the system or of the business rules that apply. Both interact to create complexity and produce unexpected outcomes.

Steps in designing technical system

Typical Technical system design

- Identify components of the system (processes, technology, infrastructure)
- Specify the problem
- Develop technical solution(s)
- Test solutions(s)
- Implement solution(s)

Socio-Technical system requires wider design assessment design – ethical impact assessment a method of addressing this

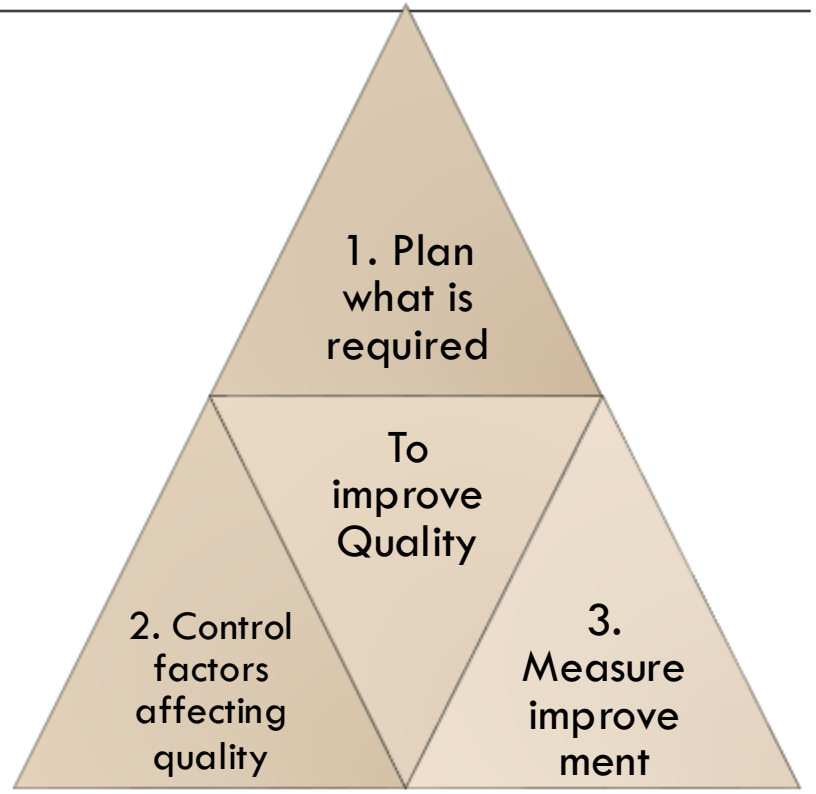


Ethical Impact design process

Ethical Impact Design Process

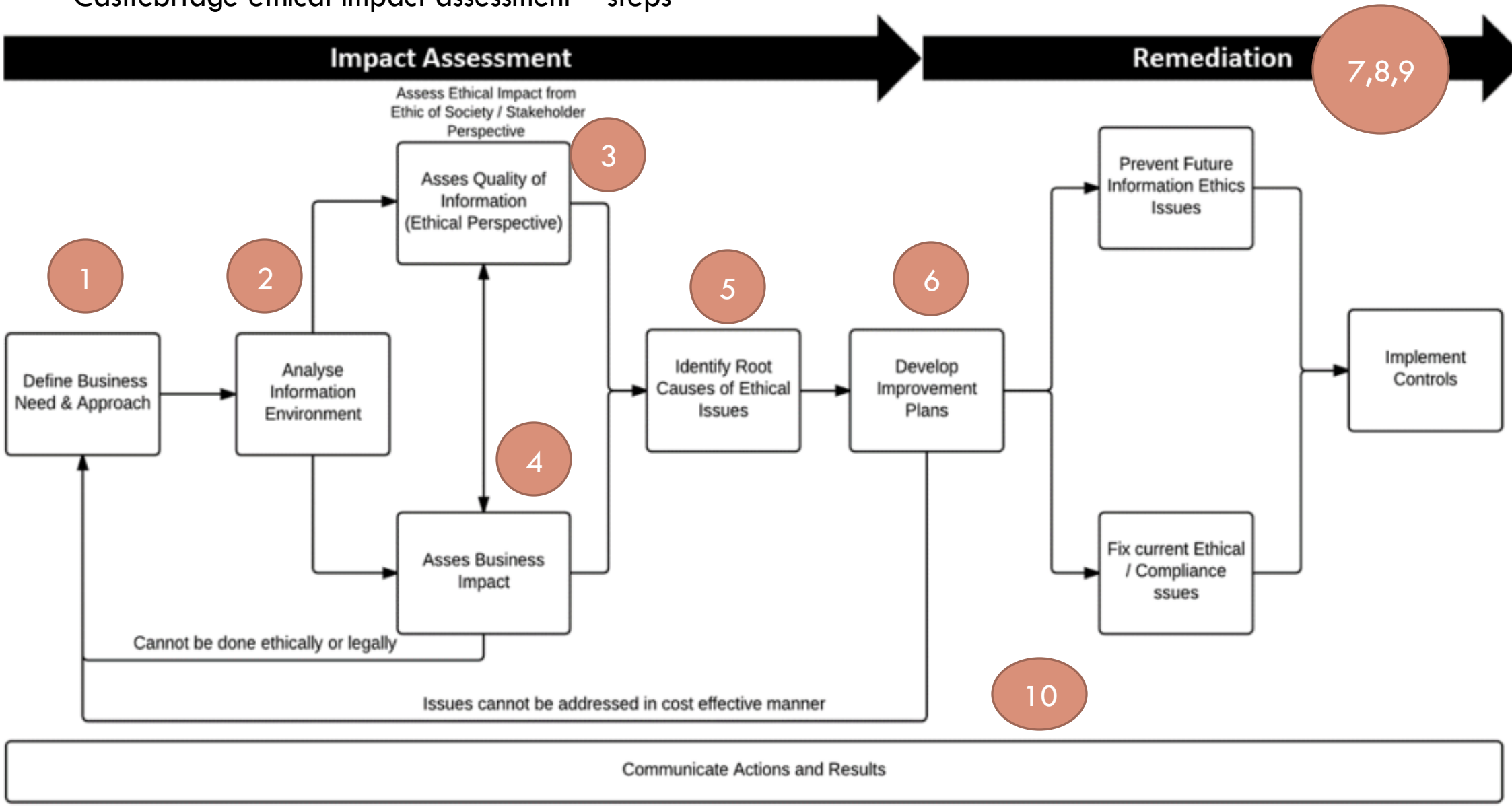
- Define business need and approach
- Analyze information environment
- Assess information privacy quality and business impact
- Identify root causes of problems
- Develop improvement plans (organizational , technological, customer/user facing, society facing)
- (Ongoing) Communication with stakeholders

From Chapter 10: O'Keefe, Katherine and O'Brien, Dara (2018). Ethical and Information Management



Juran quality assessment and improvement trilogy

Castlebridge ethical impact assessment – steps



Castlebridge ethical impact assessment method From Chapter 10: O'Keefe, Katherine and O'Brien, Dara (2018). Ethical and Information Management

Step 1: Design Business need and approach

Is the scope of your assessment clear? (focus on legal issues only, privacy issues, or broader ethical framework)

What are the goals of the review and potential conflicts?

What are the potential benefits of a review?

Who are the stakeholders associated with review?

Are you confining your review to sub-set of data processes or including all?

Use statement of business need to document your process

Example of statement of business needs

Table 10.4 Business need and approach template

We want to	Describe the information-processing activity that is the subject of the Ethical Impact Assessment. This should be sufficiently detailed to help you and your colleagues assess and identify potential ethical challenges
So that we can	Describe the organization capability that the processing is intended to provide or the social problem that the information processing is intended to address
Which will deliver the following benefits	Describe the intended benefits to the organization and to stakeholders. Focus should be on the outcomes in terms of information and process outcomes and the impact of those outcomes
To the following stakeholders	Describe the stakeholders, both internal and external, who it is intended will benefit from the proposed processing activity or whose data will be processed as part of this activity

Source: O'Keefe et all (2018)

Step 2: Analyze information environment



Social

- Is there any research on society's attitudes?
- have other organizations carried out similar projects?
- what is the internal culture of the organization?
- What are stakeholder expectations?



Technical

- what is the technology architecture?
- What are the data flows?
- How novel is the processing?
- What technology risks have been identified?
- What are the technical requirements and processes?



Legal

- What regulation rules might apply?
- What guidance is available from relevant regulations?
- What internal governance structures and rules apply?



Moral

- is it supporting or enhancing human dignity?
- Does it preserve the autonomy of the person?
- Is the processing necessary and proportionate?
- Does it uphold a common good?

Castlebridge ethical impact assessment method From Chapter 10: O'Keefe, Katherine and O'Brien, Dara (2018). Ethical and Information Management

Privacy by design questions (Moral)

Question 1: Does it preserve human dignity? Does it enhance human dignity?

Question 2 : Does it preserve the autonomy of the human?

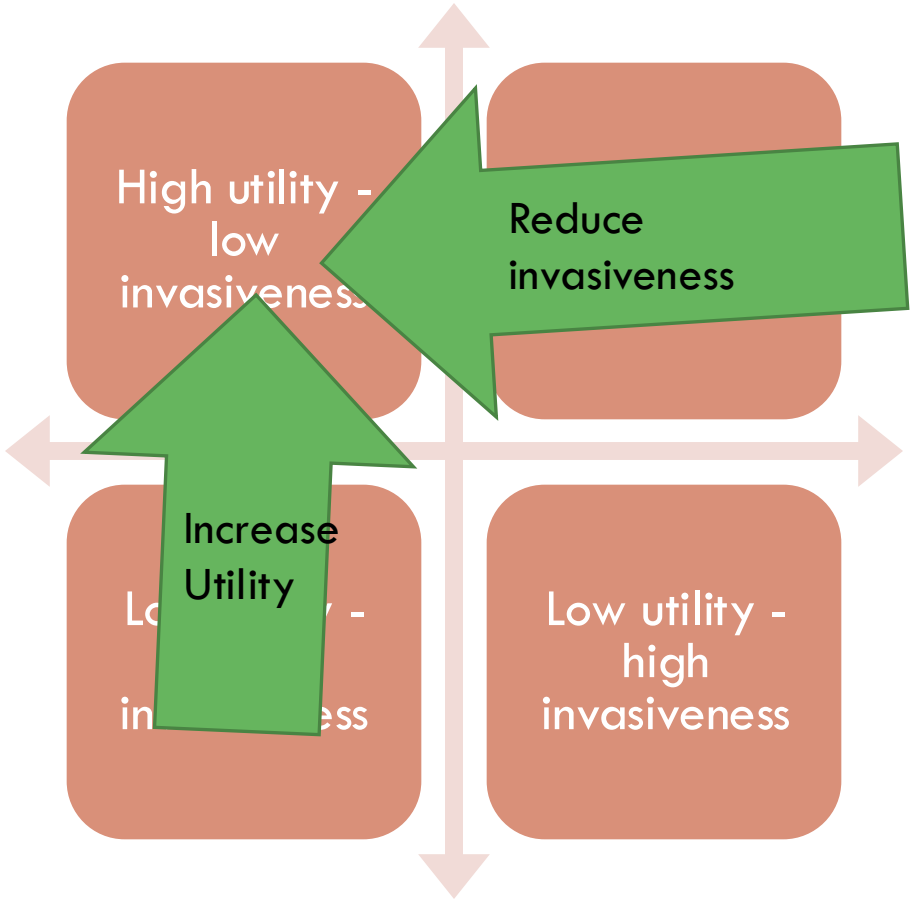
Question 3 : Is it necessary or proportionate?

Question 4 : Does it uphold the common good ? Is it of more general benefit to communities as a whole?

Steps 3 and 4: Assess information privacy quality and business impact

Examples of problematic issues

- Lack of legal basis for conducting data processing
- Proposed scope and scale of data processing not meeting the necessity of proportionality requirements under legislation
- proposed process is highly invasive and of limited utility



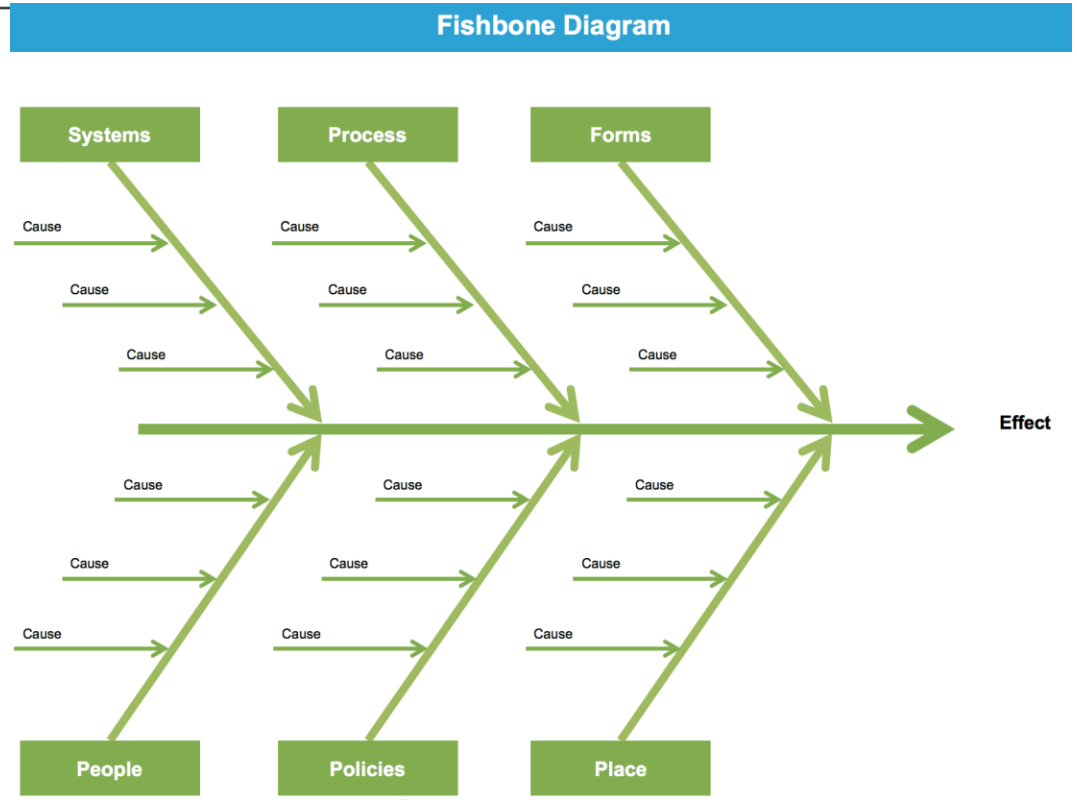
Step 5: Identify root causes of problems

Fishbone diagram to illustrate cause and effect

You write your problem statement at the head of the fish

Identify the contributing areas as the ribs

With each root cause Bing one of the ribs



Step 6: Development of improvement plans

Organizational	Technological	Customer/user facing	Society facing
• Training • Governance controls • Revised policies procedures	• Implement user access controls • Detection and logging of data access • Data masking or anonymization	• Changes to data privacy statements • Changes to how information is communicated • Controls for data subjects regarding the exercise of their rights	• Making case for change to legislation • educating mass market on benefits of processing

Step 10: communication

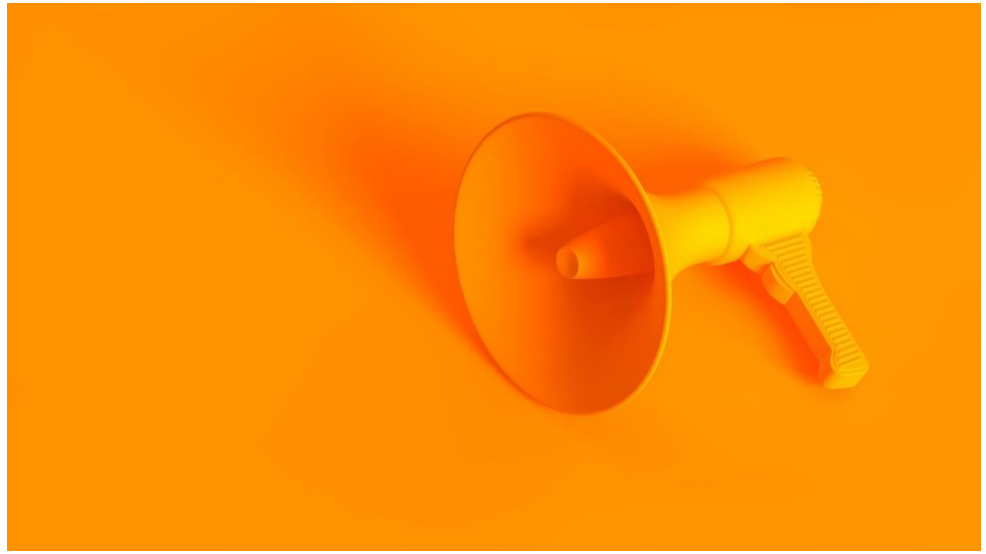
Covers entire life cycle

Who are stakeholders?

Documentation of process and steps

Clarity about steps: what is expected? what is outcome?

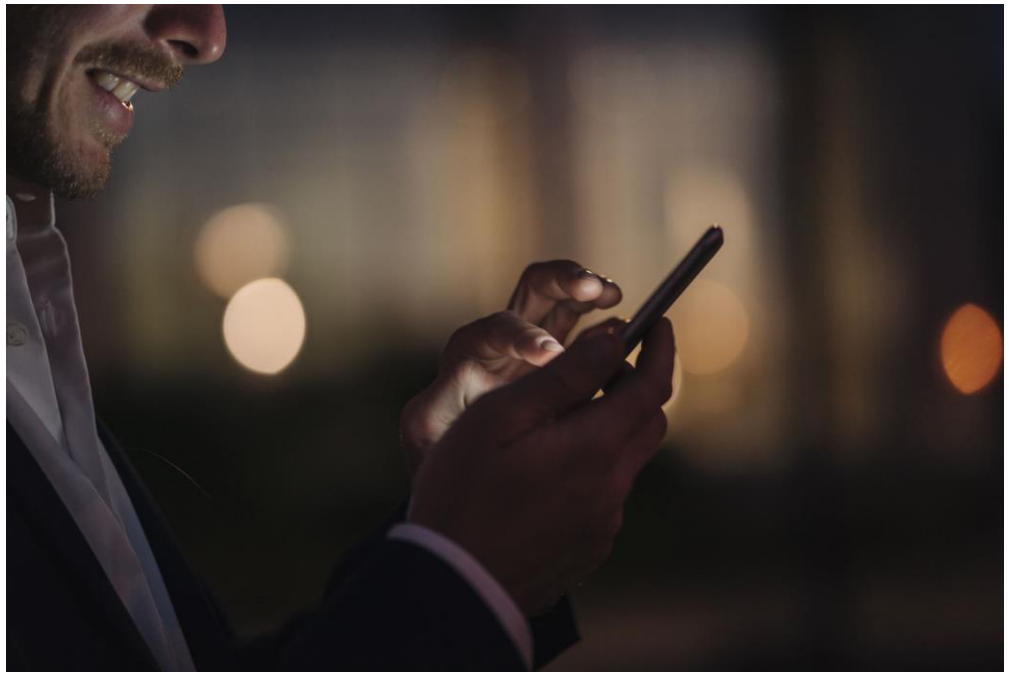
Key to alignment of ethical approaches of individuals, organisation and society



End of socio-technical systems

Questions?

Points to make?



Research Ethics

"**Research** is a formal, systematic application of the scientific approach to the study of a problem to discover new information or expand and verify existing knowledge"

Different ethical emphasis between the protection of the subject of the research and the general advancement of knowledge



Ethics Consideration

To be considered when your project falls into one scenario or both of them

Scenario 1

Human participants are involved in your study

- Requirement specification stage
- App Interface design feedback
- Evaluation and Testing stage
- Surveys and Interviews
- Etc.

Scenario 2

Secondary Dataset(s) is used in your study

- Data is/was collected by someone
- Data is taken from a web site(s)
- Data is provided by a company
- Etc.

- **Primary data** – data collected by the investigator conducting the research/work (e.g. measurements, interviews, tests)

What is a human participant?

"A living individual about whom an investigator (professional or student) conducting research obtains;

- 1.Data or
- 2.Identifiable private information

Through intervention or interaction with the individual."



Why Be Concerned About Ethics In Research?

If the objective of research is to “discover new information or expand and verify existing knowledge” it is important when conducting research involving people that this knowledge does not come at the expense of their welfare or rights.

Research ethics provide a guideline or set of principles that support researchers in conducting research so that it is done justly and without harming anyone in the process.

It is the duty of the researcher to ensure they are carrying out their research project in line with established ethical standards.

Every step of the research project, from formulating your research question to publication, needs to be informed by ethics to ensure integrity of the project.

Universal Guiding Ethical Principles

1. Respect Others' Autonomy
2. Free and informed consent;
   - Information,
   - Voluntariness and
   - Comprehension
3. Veracity
4. Respect for Vulnerable Persons
5. Privacy and Confidentiality
6. Justice and Inclusiveness
7. Consider Harms and Benefits

Minimising Harm (Non-maleficence)

Maximising Benefit (Beneficence)

Ethical Principles: Before The Research Process

What are the ethical strengths and limits of the methods?

Have the data been used ethically in a similar context before?

Are the methods respectful of your respondents' capacity and willingness to participate? Some methods will work better with some groups than with others.

Are there any potential unintended consequences of your research, e.g. disclosures of sensitive information, that may arise through those methods, potentially causing stress or embarrassment?

Do the methods proposed fit with the ethics principles mentioned earlier? If not, can the exception be justified? Is what you are proposing to do justifiable in terms of the benefits, risks and harms of your research?

Sampling

Selecting samples or groups of people to study is one of the first tasks in designing your study, and the first ethics question you face in sampling is who you include or exclude.

- You should be clear about why the people you want to study need to be involved.
- You should also reflect on who is left out of your study, and why.

Your sample will often determine which ethics committee you need to go to, and who you need to seek permission from, so it is critically important to think through the ethical implications of the sampling strategy you develop.

If your sample includes particular groups in the population, such as children and young people, or vulnerable adults (a person aged 18 years or over who may require assistance to care for themselves, or protect themselves from harm or from being exploited)

You will have additional requirements and considerations to address; for example, getting consent from guardians, as well as the participants.

Information & Consent



To recruit potential participants, you need to provide them with information about your project, in order for them to be able to give you their fully **informed consent**.

Informed consent comprises three major elements:

Information-When providing information researchers must ensure that participants are given sufficient detail about the nature of the research and the procedures involved; the information should highlight the objectives of the study, and any potential risks and benefits.

Voluntariness-Consent must be freely given and may be withdrawn at any time. Undue influence may take the form of inducement or authority over prospective participants.

Comprehension -Voluntary participation implies that participants make an informed choice while informed consent assumes that the information given is accurate. Both principles are underpinned by the principle that participant understands what it is they are being asked to participate in.

Ethical Principles: During The Research Process

Ethical issues of using group-based methods:

- Participants should be recruited ethically by providing them with the appropriate information about your research so that they can make an informed decision.
- Researchers need to be aware of the risks associated with research methods that allow for unpredictable interaction between participants or when potentially sensitive topics are being investigated.
- There is a risk of limited confidentiality when participants participate in the research process as they might accidently repeat sensitive information they learn about other participants to those not involved in the project.
- There is also a risk that participants may overshare or contribute sensitive information during the data collection process that may put them in danger of being ridiculed by their peers later on.

Managing Your Data

Dealing with data is a key element of research ethics, regardless of whether you are collecting new data, or if you are using existing data for secondary analysis or review

Data management is also increasingly highlighted by funders and research ethics committees as an explicit priority for researchers

In part this reflects the legal requirements of the Data Protection Act, but this concern is fundamentally rooted in key ethics principles, about the anonymity and confidentiality that participants are promised when they give their informed consent



Anonymising Your Data

Remove direct identifiers, e g personal information such as names and addresses

Aggregate or reduce the precision of variables that might be identifiable, e g recode postcode or neighbourhood data to Dublin North, Dublin South Galway urban, Galway rural

Generalise text variables to reduce identifiability

Restrict continuous variables to reduce outliers

Pay particular attention to anonymising relational data, for example, some anonymised variables may become identifiable when considered in combination, combining location with special educational needs

Be aware of case studies where research results have been "de-anonymized" (see case study on Moodle Sweeney, Latanya (2000) "*Simple Demographics Often Identify People Uniquely*"

Anonymising Your Qualitative Data

You do not need much of somebody's life history to work out who they are, if you know them, or if they are distinctive in some way

A distinctive event or combination of descriptions in a qualitative account could make somebody recognisable

Concerns about privacy protection can mean that qualitative data may need some editing to ensure their anonymity When the data editing is complete, researchers should take care this has not distorted their data or changes what the data is telling you, for example, deleting all possible identifiers from text or sound recordings is a simple but blunt tool that creates data that are confidential but may be unusable.

Anonymisation of qualitative data can be particularly complex, and is not simply a matter of removing personal information such as names or addresses, or of using pseudonyms

Data Storage And Security

Whether you are collecting new data or accessing existing data, you need to consider:

Your planning should take account of what you need to do with hard copies, such as, paper notes of interviews; computer files with anonymised data that are not identifiable; and computer files with personal or identifiable data.



Data Storage and Security



Files -including computer files -that contain personal or identifiable data (such as names) come under the terms of the Data Protection Act.

If your research involves data that comes under the remit of the Data Protection Act -and most research does -then it is a good idea to check with the Data Protection Officer in your organisation, to see if there are any standard protocols you should be following.

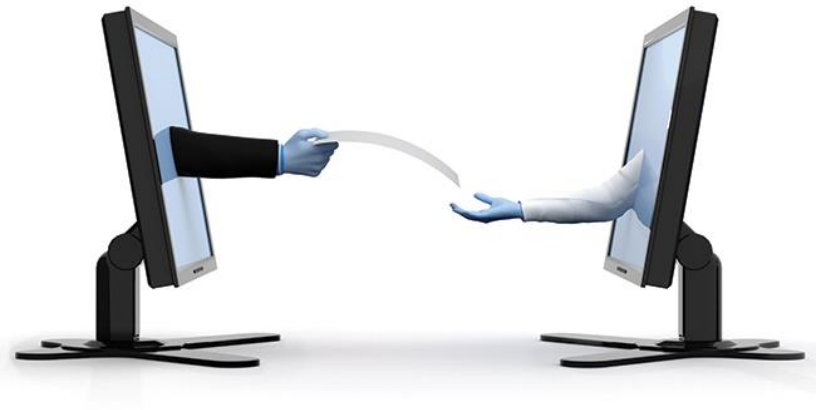
Computer files including those that have been anonymised still need to be held securely and can only be shared according to the terms of your consent from participants. Thus -for example -you need to get prior consent from participants if you plan to archive data for use by other researchers.

These files need to be encrypted or password protected, and only accessed by agreed members of the team.

Particular care needs to be taken if you are sharing files within the research team -e.g. on shared computer drives, or by email -or if you are transferring personal data beyond the research team (e.g. if a gatekeeper is giving you a list of contacts).

Ethics Principals: After Research

Think about sharing data before you begin and if there is a possibility that you will want to share the data, include that in the informed consent documents, so that you do not share what was not authorised. By the same token you have an obligation to make sure that data that is shared with you was authorised.



Data should be archived for at least five to seven years so that others can re-analyse the results, or if your findings are challenged, you can go back and look at it again.

Ethics Consideration

To be considered when your project falls into one scenario or both of them

Scenario 1

Human participants are involved in your study

- Requirement specification stage
- App Interface design feedback
- Evaluation and Testing stage
- Surveys and Interviews
- Etc.

Scenario 2

Secondary Dataset(s) is used in your study

- Data is/was collected by someone
- Data is taken from a web site(s)
- Data is provided by a company
- Etc.

- **Primary data** – data collected by the investigator conducting the research/work (e.g. measurements, interviews, tests)

Ethics Consideration

Scenario 1

Human participants are involved in your study

- Requirement specification stage
- App Interface design feedback
- Evaluation and Testing stage
- Surveys and Interviews

Declaration of Ethics Consideration Form

+ Ethics Application Form

Scenario 2

Secondary Dataset(s) is used in your study

- Data is/was collected by someone
- Data is taken from a web site(s)
- Data is provided by a company

Declaration of Ethics Consideration Form

Declaration of Ethics Consideration Form

National College of Ireland

DECLARATION OF ETHICS CONSIDERATION

School of Computing

Student Name: .....

Student ID: .....

Programme ..... **Year:** .....

Module: .....

Project Title: .....

Please circle (or highlight) as appropriate

This project involves human participants	Yes / No

Introduction

Secondary data refers to data that is collected by someone other than the current researcher. Common sources of secondary data for social science include censuses, information collected by government departments, organizational records and data originally collected for other research purposes. Primary data, by contrast, is collected by the investigator conducting the research.

A project that does not involve human participants requires ONLY completion of Declaration of Ethics Consideration Form and submission of the form on module's Moodle page

A project that involves human participants requires ethical clearance and an Ethics Application Form must be submitted through the module's Moodle page. Please refer to and ensure compliance with the ethical principles stated in NCI Ethics Form available on the Moodle page.

The following decision table will assist you in deciding if you have to complete the Declaration of Ethics Consideration Form or/and the Ethics Application Form.

Public Data	Y	Y	Y	Y	N	N	N	N
Private Data	Y	Y	N	N	Y	Y	N	N
Human Participants	Y	N	Y	N	Y	N	Y	N
Declaration of Ethics Consideration Form	x	X	x	X	X	X	x	
Ethics Application Form	X		X		X		X	

Please circle (or highlight) as appropriate

The project makes use of secondary dataset(s) created by the researcher	Yes / No
The project makes use of public secondary dataset(s)	Yes / No
The project makes use of non-public secondary dataset(s)	Yes / No
Approval letter from non-public secondary dataset(s) owner received	Yes / No

Ethics Application Form

National College of Ireland

Human Participants Ethical Review Application Form

All parts of the below form must be completed. However in certain cases where sections are not relevant to the proposed study, clearly mark NA in the box provided.

Part A: Title of Project and Contact Information

Name

Student Number (if applicable)

Email

Status:

Undergraduate ☐
Postgraduate ☐
Staff ☐

Supervisor (if applicable)

Title of Research Project

Category into which the proposed research falls (see guidelines)

Research Category A ☐
Research Category B ☐
Research Category C ☐

Have you read the NCI Ethical Guidelines for Research with Human Participants?

Yes ☐
No ☐

Please indicate any other ethical guidelines or codes of conduct you have consulted

Has this research been submitted to any other research ethics committee?

Yes ☐
No ☐

If yes please provide details, and the outcomes of this process, if applicable:

Ethics Consideration

Scenario 2

Secondary Dataset(s) is used in your study

Scenario 2 A

PUBLIC Secondary Dataset(s) is used

- *A letter/email from the copyright holder is required to confirm potential use of data*
- *OR, Citation and link to the web site where permission is granted must be provided*

Scenario 2 B

PRIVATE Secondary Dataset(s) is used

- *Permission to use the dataset MUST BE received from the owner*
- *An approval letter/email from the owner must be provided*

REMEMBER!

It is students' responsibility to ensure that they have the correct permissions/authorizations to use any data in a study.

Ethics Application Form

High Ethical Risk - If your work involves any of the following:

- Vulnerable groups
- Sensitive topics
- Risk of psychological or mental distress
- Risk of physical stress or discomfort
- Any other risk to participants
- Use of drugs or invasive procedures (e.g. blood sampling)
- Deception or withholding of information from participants
- Conflict of interest issues
- Access to data by individuals or organisations other than the researchers
- Any other ethical dilemmas

- **Vulnerable Groups**
  - Children (under 18 years of age)
  - The older old (aged 85+)
  - People with an intellectual or learning disability
  - Individuals or groups receiving help through the voluntary sector
  - Those in a subordinate position to the researcher (e.g. employees)
  - Any other groups who might not understand the research and consent process

If you work directly with children, Garda Vetting is required in advance of the Ethics Form submission.